

AI Use Declaration Form

Course: Introduction to Artificial Intelligence

Lab Title: Lab 1 Parts A & B — Uninformed and Informed Search Algorithms

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GitHub Repository Link: https://github.com/Joram-Hanson/Joram_Hanson_Lab1.git

Date Submitted: 20/06/2026

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	Yes
Estimated percentage of the work influenced by AI	15 %
Did you attach evidence of AI use where applicable?	Yes

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
Claude (Anthropic)	To help me understand lecture concepts (clarify what each lab section was asking for, and explain error messages I ran into while coding.	Questions such as “why does BFS use a FIFO queue”, “explain what TODO 2 is asking for”, and pasting error tracebacks to ask why my code was failing.	Conceptual explanations of search algorithm behaviour, and pointing out bugs in code I had already written (e.g. a typo in a function name, wrong indentation).	I wrote the TODO implementations myself, ran every cell, and checked the output against the self-checks and expected behaviour described in the lab before moving on. Where Claude pointed out a bug, I located the line myself and fixed it rather than copying a

				replacement block.

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	Attached? (Yes, No, N/A)	File Name
AI-generated draft	N/A	
Prompt history	Yes	AI Use Declaration Form.docx
Screenshot of AI interaction	Yes	AI Use Declaration Form.docx
Exported AI conversation	N/A	
AI-generated code snippet	Yes	AI Use Declaration Form.docx
Revised version showing student input	N/A	

4. Student Declaration

I declare that:

Declaration Statement	Response (Yes/No)
The submitted work is my own work.	Yes
Any use of AI tools has been clearly declared in this form.	Yes
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	Yes
I reviewed, tested, edited, and improved any AI-generated content before submission.	Yes
My AI usage does not exceed 25% of the entire work.	Yes
I understand that undeclared or excessive AI use may be treated as academic misconduct.	Yes

Student Signature: __ J.E.H. _____

Date: __ 20/06/2028 _____

SCREENSHOTS

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framework does everything else.

Section 8 — A Search*
Also a one-liner. A* defines $f(n) = g(n) + h(n)$. The $g(n)$ part is already stored on the node.

Section 9 — Uniform-Cost Search
Fully provided. It's just A* with $h(n) = 0$, included as a baseline to show how much work the heuristic actually saves.

Section 10 — Weighted A*
One small TODO. Weighted A* defines $f(n) = g(n) + w \times h(n)$. The weight w makes the search greedier — higher w means faster but less optimal.

Section 11 — Run on the Sample Map
No TODOs — just run the algorithms on the familiar Part A map and study the results table. Focus on comparing nodes expanded between UCS and A*, and solution cost between A* and Greedy.

Section 12 — Turbulence Map
Also provided and ready to run. This is where Greedy's weakness gets exposed — a band of expensive terrain sits between start and goal. Predict the two path costs before you run it, then check if you were right.

Section 13 — Inadmissible Heuristic Experiment
One TODO — writing an *inflated heuristic* — a function that multiplies Manhattan distance by a factor. Then you run A* with it and observe what breaks. Write your observations in the

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Can you explain each section of Part B and what it is asking me to do? Please break it down in simple terms and explain the purpose of each section.

Section 0 — OOP Design Map
This is just a reference table showing how the concepts from your lectures map onto the Python code you'll write. Pay attention to two key changes from Part A: *reached* becomes a dictionary instead of a set, and the goal test happens at *pop* time not *generation* time. Understanding why these changed is important for the reflection questions.

Section 1 — Imports
Already covered these — same as Part A plus `heapq`, `itertools`, and `math` for the priority queue and distance calculations.

Section 2 — Foundations from Part A
Everything here is provided for you. It's a copy of `Problem`, `Node`, `SearchResult`, the helper functions, `GridProblem`, `expand`, and `plot_path`. You don't write anything here — just run the cells so everything is available for the rest of the notebook.

Section 3 — WeightedGridProblem
Your first TODO. This extends `GridProblem` with terrain costs — instead of every move costing 1, entering a cell now costs whatever that cell's terrain value is. You only need to

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```
current_stack_size -= 1

if cutoff_occurred:
    return (None, "cutoff")
else:
    return (None, "failure")
```

for this TODO section 11 in Part A how should i go about it

Show less

11:48 PM

Just to clarify — this is actually **TODO 8** (`_recursive_dls`), not TODO 10/11. TODOs 10 and 11 are the custom maps at the end.

That said, I can see you've already written an implementation here. Rather than starting fresh, have you actually **run it yet**?

If it's working correctly on the sample map, the best way to check is to look at a few things in your results table:

- Does DLS return `"success"` when the limit is high enough?
- Does it return `"cutoff"` when the limit is too small to reach the goal?
- Does the path it finds actually connect start $\downarrow$ goal without passing through obstacles?

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```
import numpy as np
```

— Used for converting your grid into an array for the visualisation.

```
import pandas as pd
```

— Used to display your results as a neat comparison table.

```
import matplotlib.pyplot as plt
```

and `matplotlib.patches` — These handle the grid visualisation, drawing coloured rectangles for obstacles, paths, start, and goal.

Extra Imports in Part B Only

```
import heapq
```

— The engine behind the priority queue. Informed search always needs to expand the node with the lowest `f(n)` value, and `heapq` keeps your list sorted so the smallest is always first.

```
import itertools
```

— Used specifically for `itertools.count()`, which generates a unique number for each node pushed into the priority queue. This acts as a tie-breaker when two nodes have the same `f(n)`.

```
import math
```

— Needed for `math.sqrt()` in the Euclidean distance heuristic, which didn't exist in Part A.

The Setup Sections

Beyond imports, the early sections set up the building blocks everything else depends on — the abstract `Problem` interface, the `Node` and `SearchResult` dataclasses, and the helper functions like `reconstruct_path`. None of these $\downarrow$ any searching themselves; they're the foundation that your search algorithms sit on top of.

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What are the setup sections and imports in this assignment? Can you explain them in simple terms and tell me what they are used for?

Sure! Both parts share most of the same imports with Part B adding a few extras. Here's a simple breakdown:

Imports in Both Parts

from abc import ABC, abstractmethod

 — This gives you the tools to create "template" classes. Think of it like a contract: if you say a class is abstract, any class that inherits from it *must* fill in certain methods. That's how `Problem` and `SearchAlgorithm` enforce that every subclass provides things like `actions()` and `is_goal()`.

from dataclasses import dataclass

 — A shortcut for creating clean data-holding classes. Instead of writing a full `__init__` method manually, you just list the fields. `Node` and `SearchResult` use this.

from collections import deque

 — Gives you a double-ended queue, which is the data structure BFS needs. It lets you add to one end and remove from the other efficiently.

from typing import Any, List, Tuple, Optional, Dict, Iterable

 — These are just type hints. They don't change how the code runs but make it clearer what kind of data each function expects and returns.

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The OOP Design

Both parts follow the same structure: a `Problem` class knows the environment, a `Node` class tracks search tree info, and `SearchAlgorithm` classes know how to search. Keeping these separate is the whole point — the same search algorithm should work on any problem, not just drone grids.

Key Guidelines to Keep in Mind

- Complete all `TODO` cells yourself
- Create at least two custom maps per part and interpret the results
- Answer all reflection questions in your own words
- Write an AI Use Declaration — what you used AI for, which parts it touched, and how you verified or changed anything it produced
- AI use must stay under 25% of the total work
- Submit via a public GitHub repo, not just a file upload

What would you like to dig into first?

What are the setup sections and imports in this assignment? Can you explain them in simple terms and tell me what they are used for?

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Can you give me a brief overview of the assignment and the main guidelines I should follow while completing it?

Sure! Here's a quick overview based on both notebooks:

The Big Picture

You're building pathfinding logic for an autonomous drone doing environmental monitoring — detecting things like illegal mining activity. The drone needs to navigate a grid from a start point to a goal while avoiding obstacles.

Part A focuses on *uninformed* search — the drone searches blindly with no sense of direction:

- BFS, DFS, Depth-Limited Search, and Iterative Deepening Search
- Every move costs the same (1 step = 1 cost)

Part B focuses on *informed* search — the drone now has GPS and knows terrain costs:

- Greedy Best-First, A*, Weighted A*, and Uniform-Cost Search
- Terrain matters now (forest, water, restricted zones all cost differently)

The OOP Design

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For this assignment, please act as a learning guide rather than a coding assistant. Help me understand concepts, explain errors I encounter, and provide hints or guidance when needed. Please do not provide code implementations or complete solutions—I want to write and figure out the code myself.

Got it! I'll be your concept guide and error explainer throughout this assignment — helping you understand the *why* behind things, pointing you in the right direction when you're stuck, and letting you work through the actual code yourself.

Whenever you hit an error or want to understand something better, just paste what you're seeing and I'll help you make sense of it. What would you like to start with?

Can you give me a brief overview of the assignment and the main guidelines I should follow while completing it?

Sure! Here's a quick overview based on both notebooks:

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the goal

- The cost-5 band across row 11 is visible
- Start (red) is at top-left, goal (green) is inside the box — good design

One thing to think about:

Your title says "*Greedy fell for the trap*" — but looking at the path, Greedy actually *reached* the goal successfully through the bottom opening. The trap concept works when Greedy wastes time entering the pocket from the wrong side before finding the opening.

The more interesting question to ask yourself is: **does Greedy's path cost more than A*'s path on this map?** That's what your results table should show. If Greedy found the goal but with a higher cost or more nodes expanded than A*, then the trap is working as intended.

For your interpretation paragraph, the key things to address are:

- Which algorithm found the cheaper path and why
- Whether Greedy was misled by only looking at $h(n)$
- What role the cost-5 band played, if any, in the results

Have you plotted the A* path on the same map to compare them side by side?

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